Causal Inference Library: Why

Ain Shams University, Faculty of Engineering

Ahmed Abdulmaksoud, Ahmed Bakry, Ahmed Emad, Emad Mostafa, Kareem Elsawah, Mohamed Kotb

***Abstract*—In this paper, we present "Why", a causal inference library for structural causal modeling and identification. "Why" implements a variety of algorithms, including the pc algorithm for causal discovery and multiple independence tests such as pearsons correlation, partial correlation, and chi squared. We also introduce the use of Generative Neural Networks (GNNs) and Causal Generative Neural Networks (CGNNs) for orienting the edges of structural causal models. Additionally, "Why" offers bound estimation for the average treatment effect (ATE) under various assumptions, as well as sensitivity analysis for the presence of unobserved linear confounders. Our library also includes implementations of backdoor adjustment and ATE estimation using the COM, GCOM, and TARNet methods. Overall, "Why" provides a comprehensive toolkit for causal inference in a range of settings. The library can be found at https://github.com/MRSAIL-Mini-Robotics-Software-AI-Lab/Why**

I. Introduction

Causal inference is the process of determining the relationship between an intervention or treatment and its effect on an outcome. It is a central problem in many fields, including economics, sociology, psychology, and medicine, as understanding causality allows us to make informed decisions, design effective interventions, and predict the consequences of our actions. However, accurately determining causality is often technically challenging, as it requires the ability to effectively control for confounding variables and consider alternative explanations for observed relationships. In particular, there are several technical challenges that must be addressed in order to accurately perform causal inference, including:

- Selection bias: This occurs when the treatment assignment process is not random, which can lead to non-comparable treatment and control groups.
- Confounding variables: These are variables that are correlated with both the treatment and the outcome, and can produce a spurious relationship between the treatment and outcome if not properly accounted for.
- Hidden confounding variables: These are variables that are not measured or observed, but may still influence both the treatment and the outcome.
- Temporal precedence: It is necessary to establish that the treatment occurred before the outcome in order to infer causality.
- Non-random treatment assignment: If the treatment is not randomly assigned, it may be difficult to disentangle the treatment effect from other factors that may influence the outcome.
- Generalizability: It may be difficult to determine the extent to which the results of a causal inference study can be generalized to other populations or settings.

There are many different methods and techniques that have been developed for causal inference, ranging from traditional statistical techniques such as regression analysis to more modern machine learning approaches like causal deep learning. Some commonly used methods include:

- Randomized controlled trials (RCTs): These are experiments in which subjects are randomly assigned to treatment and control groups, allowing for a clean separation of the treatment effect from other confounding factors. RCTs are widely considered the "gold standard" for causal inference, but they may not be practical or ethical in all situations.
- Observational studies: These are studies in which the treatment is not randomly assigned, and therefore the separation of the treatment effect from other confounding factors is more challenging. Observational studies may use statistical techniques such as propensity score matching or instrumental variables to try to control for confounding.
- Matching: This involves finding pairs of individuals with similar characteristics and comparing the outcomes for those who received the treatment versus those who did not.
- Instrumental variables: This involves using an external factor that influences the treatment assignment but is not directly related to the outcome, in order to infer the causal effect of the treatment.

While these methods can be effective in certain situations, they also have limitations and may not be applicable in all cases. For example, RCTs may not be feasible or ethical in certain situations, and observational studies may be subject to uncontrolled confounding. Additionally, many methods rely on strong assumptions that may not hold in practice, such as the assumption of no unobserved confounders or the assumption of linear relationships. As a result, there is ongoing research into developing new and improved methods for causal inference.

This work focuses on the problem of causal inference in observational studies, in which the treatment is not randomly assigned. In such situations, accurately determining the causal effect of the treatment can be particularly challenging, as it requires controlling for confounding variables and considering alternative explanations for the observed relationships. To

address these challenges, we have implemented a range of methods for causal inference in observational studies which we discuss in the following section. These methods can be broadly divided into:

- Causal discovery: These are methods for identifying the structure of a causal model, including the variables and their relationships, from observational data. The pc algorithm is a well-known method for causal discovery that we have implemented in the "Why" library.
- Bound estimation: These are methods for estimating the bounds of the ATE under various assumptions, such as no assumptions bounds, monotone treatment response, optimal treatment selection, and monotone treatment selection.
- Sensitivity analysis: These are methods for evaluating the robustness of causal estimates to the presence of unobserved confounders.
- Backdoor adjustment: These are methods for obtaining adjustment sets for ATE estimation, which can help control for confounding variables.
- ATE estimation: These are methods for estimating the ATE, or the average effect of the treatment on the outcome. We have implemented several ATE estimation methods, including COM, GCOM, and TARNet.

II. Causal Discovery

Causal discovery is the process of identifying the causal relationships between variables in a system. In causal inference, the goal is to use observational data to draw conclusions about the causal effects of different variables on each other. Causal discovery algorithms can help to identify the underlying structure of a system and reveal the potential causal relationships between variables. This can be useful for understanding the underlying mechanisms at play in a system, predicting how changes to one variable might affect other variables, and for designing interventions that can have a desired effect on the system.

There are several approaches to causal discovery, including graphical methods, constraint-based methods, and score-based methods.

- Graphical methods involve constructing a graphical representation of the relationships between variables, where the nodes in the graph represent variables and the edges represent the relationships between them. The structure of the graph can reveal potential causal relationships between the variables.
- Constraint-based methods make use of statistical dependencies between variables to identify potential causal relationships. These methods involve identifying the minimal set of statistical dependencies that are sufficient to imply all of the dependencies in the data.
- Score-based methods involve defining a score function that measures the strength of the relationship between two variables, and then using this score function to identify the strongest relationships in the data.

In general, the goal of causal discovery is to identify the underlying structure of a system and the causal relationships between its variables, without making any strong assumptions about the form of these relationships. This can be a challenging task, as real-world systems can be very complex and may involve many variables that are interdependent in intricate ways.

In our analysis, we are particularly interested in methods that use graphical or constraint-based approaches. One specific algorithm we are interested in is the PC (Peter and Clark) Algorithm, which employs conditional independence tests to construct a Bayesian network from a set of data. We begin by briefly describing graphical and constraint-based approaches, then onto the independence tests implemented. Finally, we present the PC algorithm in more details.

A. Graphical Methods

Graphical methods are a type of approach to causal discovery that involves constructing a graphical representation of the relationships between variables. The nodes in the graph represent variables, and the edges represent the relationships between them. The structure of the graph can reveal potential causal relationships between the variables.

There are several different types of graphs that can be used in causal discovery, including directed acyclic graphs (DAGs), semi-Markov models, and Bayesian networks.

Directed acyclic graphs (DAGs) are a type of graphical model that represent the causal relationships between variables as directed edges. A directed edge from variable A to variable B indicates that A has a direct causal effect on B. DAGs can be used to represent both linear and nonlinear relationships between variables, and can be used to identify the potential causal relationships between variables in a system.

B. Constraint-based Approach

Constraint-based approaches for causal discovery aim to identify the causal relationships between variables in a system by examining the constraints that these relationships impose on the distribution of the variables. These methods rely on the assumption that if two variables are causally related, then there should be certain constraints on the distribution of one variable given the other.

One popular constraint-based method for causal discovery is the PC Algorithm, which was mentioned earlier. This algorithm works by iteratively adding edges to a graph, with the goal of representing the dependencies among the variables in the data. It makes use of conditional independence tests to determine whether or not to add an edge between two variables. If the variables are found to be independent, then no edge is added. If they are dependent, then an edge is added to represent the dependency.

Another constraint-based method is the FCI (Fast Causal Inference) Algorithm. This algorithm uses the d-separation criterion to identify conditional independencies in the data, which can then be used to infer the presence of causal relationships. It works by constructing a graph from the data and

then checking for d-separation in the graph, which indicates the presence of a causal relationship between the variables. If a causal relationship is detected, then an edge is added to the graph to represent the relationship. These methods can be used to identify the causal structure of a system, which can then be used to make predictions or perform other types of analysis.

C. Independence Tests

Independence tests are statistical tests that are used to determine whether two variables are independent of each other. In the context of causal discovery, independence tests can be used to determine whether there is a causal relationship between two variables.

The idea behind independence tests is that, if two variables are truly independent, then the value of one variable should not affect the distribution of the other variable. Conversely, if two variables are dependent, then the value of one variable is likely to affect the distribution of the other variable. Independence tests can be used to determine whether the observed dependence between two variables is statistically significant, or whether it could have arisen by chance.

There are several different independence tests that can be used in causal discovery, including chi-square tests, and partial correlation tests. These tests each have their own assumptions and limitations, and it is important to choose the appropriate test based on the characteristics of the data and the research question being addressed.

1) Conditional Chi-squared Test: The chi-square test is a statistical test that is used to determine whether there is a significant association between two categorical variables. It is a nonparametric test, which means that it does not assume that the data follows a particular distribution.

To perform a chi-square test, the observed frequencies of the categories for each variable are compared to the expected frequencies under the assumption of independence. The chi-square statistic is calculated as the sum of the squared differences between the observed and expected frequencies, divided by the expected frequencies. If the observed frequencies are significantly different from the expected frequencies, then it can be concluded that there is a significant association between the two variables.

To condition on a set of variables, we derive the contingency table between the two target variables for each of the unique combinations of the conditioning variables then carry out the chi-squared test on each table. For the target variables to be independent, all the contingency tables must pass the chi-squared test. A problem with this approach is the need for enough data to cover all unique combinations of the conditioning variables.

The chi-square test is sensitive to sample size, and the p-value for the test will be larger for larger sample sizes. This means that it is more likely to find a significant association between two variables when the sample size is large, even if the association is actually weak. It is therefore important to consider the sample size when interpreting the results of a chi-square test.

The chi-square test is a commonly used independence test in causal discovery, particularly for categorical data. This calculation can be done in O(n) time, where n is the number of observations in the data. It is easy to compute and does not require the assumption of normality for the data. However, it does have some limitations, including the sensitivity to sample size and the fact that it is only suitable for categorical data.

2) Partial Correlation Test: The Partial Correlation test is a statistical test that is used to determine whether two continuous variables are conditionally independent of each other given the values of one or more other variables. It is based on the Pearson correlation coefficient, which measures the linear relationship between two variables.

To perform the Pearson's conditional independence test, the following steps are typically followed:

1) Collect the data: The first step is to collect a sample of data for the two variables that are being tested for conditional independence, as well as the other variables that are being used to control for the effects of these variables.
2) Estimate the conditional correlation: The Pearson correlation coefficient is calculated between the two variables being tested, while controlling for the effects of the other variables. This can be done using a partial correlation analysis or a multivariate regression model.
3) Calculate the p-value: The p-value for the test is calculated using the Pearson correlation coefficient and the sample size. The p-value represents the probability of obtaining a Pearson correlation coefficient as extreme as the one observed in the sample data, given that the null hypothesis is true.
4) Decide on the significance level: The significance level is the threshold that is used to determine whether the observed correlation is statistically significant. Typically, a significance level of 0.05 is used, which means that the observed correlation is considered statistically significant if the p-value is less than 0.05.
5) Interpret the results: If the p-value is less than the significance level, then the null hypothesis is rejected and it can be concluded that there is a significant conditional correlation between the two variables, and that they are not conditionally independent given the values of the other variables. If the p-value is greater than the significance level, then the null hypothesis is not rejected, and it can be concluded that there is no significant conditional correlation between the two variables, and that they are conditionally independent given the values of the other variables.

To perform a partial correlation analysis mentioned in step (2), the Pearson correlation coefficient is calculated between each pair of variables while controlling for the effects of the other variables. This can be done using the following steps:

1) Calculate the Pearson correlation coefficient between each pair of variables using the formula

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{(n\sum x^2 - \sum x^2)(n\sum y^2 - \sum y^2)}}$$

where x and y are the two variables being tested, n is the number of data points, and sum(xy) is the sum of the products of the values of x and y.

2) Regress each variable on the other variables: For each variable, run a regression analysis to predict that variable using the other variables as predictors.
3) Calculate the residuals: Calculate the residuals for each regression analysis, which represent the differences between the observed values of the dependent variable and the predicted values.
4) Calculate the partial correlation coefficient: Calculate the Pearson correlation coefficient between the residuals for each regression analysis. This partial correlation coefficient represents the correlation between the two variables while controlling for the effects of the other variables.

Overall, the goal of estimating the conditional correlation between the two variables being tested is to determine whether there is a significant relationship between the two variables given the values of the other variables and to quantify the strength of this relationship. This can be done using either a partial correlation analysis or a multivariate regression model, depending on the specific needs of the analysis.

D. PC Algorithm

PC Algorithm, also known as Peter and Clark algorithm, is a set of rules used to infer causal relationships among variables in a given data set. This algorithm can be used to identify which variables are independent and how each variable has an effect on the others. It utilizes Pearson correlations and partial correlations between different variables that can help distinguish direct and indirect causes from confounding factors. This type of data analysis can be useful when analyzing large data sets to determine the real causes of certain phenomena. The PC Algorithm is designed to identify all possible causal relationships in a data set, yet it does not guarantee that these results are absolutely true. Nevertheless, its results provide valuable insight into what behavior can be attributed directly to certain variables.

At a high level, the PC Algorithm works by analyzing a dataset and generating an undirected graph representation of this data. This graph contains nodes which represent variables in the dataset and edges that connect two nodes if they have some sort of statistically significant dependency. Then, it uses this graph representation to search for patterns which indicate potential causal relationships between variables.

Ultimately, after running through all possible permutations of a given dataset, the PC Algorithm can give us an overview of what it considers most likely causal relationships to be within that data set. It does so by creating an oriented graph showing directions across variables and assessing each possible relationship according to its statistic strength metric such as CI (conditional independence). From there we can analyze our result further to understand more precisely what causes what given our dataset context and assumptions made during setup.

1) Identifying the skeleton: The first step in PC algorithm is to the identify the main skeleton of the graph. It starts by form a complete undirected graph with all the features.

Theorem 1 . If vertices X and Y are not adjacent in a DAG G, then there is a set of vertices Z which is either a subset of adj(X,G)/Y or a subset of adj(Y,G)/X such that Z d-separates X and Y in G.

Based on Theorem 1, For each edge, the PC algorithm tests if the pair of variables connected by the edge, X and Y , are independent conditioning on a subset Z of all neighbours of X and Y . The tests are organised by levels (based on the size of the conditioning sets, e.g. the depth d).

At the first level (d = 0), all pairs of vertices are tested conditioning on the empty set. Some of the edges would be deleted and the algorithm only tests the remaining edges in the next level (d = 1). The size of the conditioning set, d, is progressively increased (by one) at each new level until d is greater than the size of the adjacent sets of the testing vertices. The number of conditional independence tests is small when the true graph is sparse. If the conditional independence tests are truly correct, then the PC algorithm will return the true graph.

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2) Orientation based on immoralities: For each triple of variables (A, B, C) such that A and B are adjacent, B and C are adjacent, and A and C are not adjacent, orient the edges A - B - C as $A \rightarrow B \rightarrow C$, if B was not in the set conditioning on which A and C became independent and the edge between them was accordingly eliminated. We call such a triple of variables a v-structure.

3) Orientation based on colliders: The main idea is to use the fact that we discovered all immoralities to orient more edges. For each triple of variables such that $A \rightarrow B$ - C, and A and C are not adjacent, orient the edge B - C as $B \rightarrow C$. This is called orientation propagation.

E. CGNN: Causal Generative Neural Network

CGNNs (Causal Generative Neural Networks) are a modern method for determining the edges in a structural causal model (SCM), which is a graphical representation of the causal relationships between variables, with edges representing direct causal influences and nodes representing the variables. Determining the edges in an SCM is a crucial step in causal

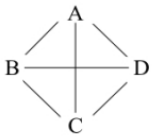
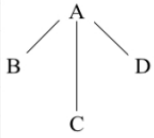
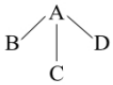
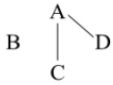
Level	Graph	#CI tests	Test	Result	Updated Graph
1		1	I(A, B)?	No	
		2	I(A, C)?	No	
		3	I(A, D)?	No	
		4	I(B, A)?	No	
		5	I(B, C)?	Yes	
		6	I(B, D)?	Yes	
		7	I(C, A)?	No	
		8	I(C, D)?	Yes	
		9	I(D, A)?	No	
2		10	I(A, B | C)?	Yes	
		11	I(A, C | D)?	No	
		12	I(A, D | C)?	No	

Fig. 1: Example of generating the skeleton

inference, as it allows us to understand the relationships between the variables and make predictions about their behavior. However, traditional methods such as constraint-based and independence test-based approaches can struggle to distinguish between chains and forks, which fall under the same Markov equivalence class and cannot be distinguished based on statistical relationships alone.

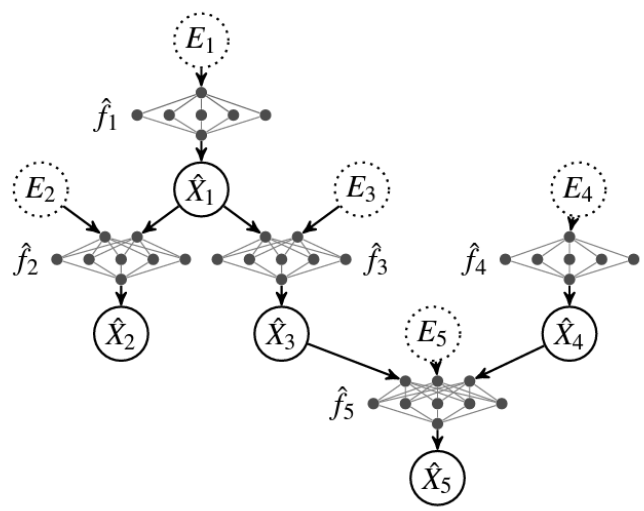
To further understand the process of using GNNs to determine the direction of edges in a structural causal model (SCM), it is helpful to consider the specifics of how these models are trained. GNNs are trained on observational data, which means that they are not based on randomized controlled experiments and may be subject to confounding variables. To evaluate the direction of a single edge, a small neural network with a single hidden layer and ReLU activations is fitted on both directions of the edge using the Maximum Mean Discrepancy (MMD) loss. The MMD loss is a non-parametric measure that compares the statistical properties of two data samples and is often used as a replacement for the KL divergence in cases where the underlying distributions are unknown or difficult to estimate. The use of the MMD loss allows the GNN to make inferences about the direction of the edge based on the relationships between the variables in the data, rather than relying on assumptions about the distributions or the functional form of the relationships.

In addition to the MMD loss, the GNN also takes a random noise variable as input. This noise variable serves as a regularization term that helps to prevent overfitting and improve the generalizability of the model. By incorporating this noise term, the GNN is able to learn more robust and reliable relationships between the variables, rather than just memorizing the patterns in the training data. After fitting the small neural network on both directions of the edge using the MMD loss and the noise variable, this process is repeated multiple times, and the losses are averaged to determine the direction of the edge. This averaging step helps to reduce the variance of the estimates and improve the stability of the inferences.

Causal Generative Neural Networks (CGNNs) are an extension of GNNs (Generative Neural Networks) that allow for the global evaluation of the edges in a structural causal model (SCM). Whereas GNNs are used to locally evaluate the direction of a single edge, CGNNs are designed to consider the entire SCM as a whole and determine the optimal orientation of all the edges based on the relationships in the data. To accomplish this, CGNNs use an "autoregressive" architecture, which means that each variable in the SCM is modeled as a function of the other variables in the model. This allows the CGNN to capture the interdependencies between the variables and make more accurate inferences about the relationships between them.

A schematic of the CGNN architecture is shown in figure 2. As can be seen, the CGNN consists of a series of GNNs, one for each causal mechanism in the SCM. The input to each GNN is the data for the corresponding variable, along with the noise term. The output of the GNN is the estimated value of the variable, which is passed to the next GNN in the chain. This process continues until all the variables in the SCM have been modeled.

Fig. 2: CGNN Architecture



To fit the CGNN, the model is trained on observational data using the MMD loss, as described in the previous paragraph. The model is fit multiple times, and the losses are averaged to evaluate the performance of the CGNN on the entire SCM. A hill climbing algorithm is then used to search for changes in edges that result in a lower loss. This process can help to improve the global orientation of edges in the SCM, and ultimately leads to more accurate and reliable causal inferences.

One of the key advantages of CGNNs is that they are able to handle high-dimensional and noisy data, which can be a challenge for traditional causal inference methods. By using a neural network architecture, CGNNs are able to extract useful information from the data and make robust inferences about the relationships between the variables. This is particularly important in cases where the data is complex or incomplete, as it allows the CGNN to account for uncertainty and make more reliable predictions.

In summary, CGNNs are a powerful tool for determining the edges in a structural causal model. By combining the strengths of GNNs and an autoregressive architecture, CGNNs are able to globally evaluate the orientation of edges in an SCM and make more accurate and reliable causal inferences. This is particularly useful in situations where the data is high-dimensional or noisy, as CGNNs are able to extract useful information from the data and make robust predictions despite these challenges.

III. Observational Studies

A. Backdoor Adjustment

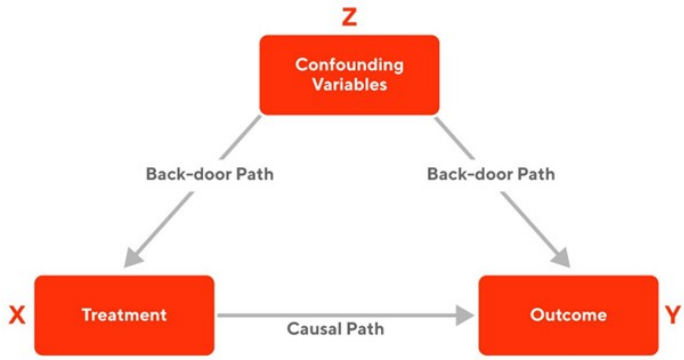
Back-door adjustment is a causal analysis algorithm used to measure the causal impact between a treatment and the outcome. Since we only need to measure the causal association, we need to adjust for all observed confounders to eliminate any non-causal association.

To understand this concept lets discuss a simple example of food ordering application (Talabat). Let's say the development team worked on a bug in the application and was able to fix the bug. The next quarter the number of orders Talabat has received was doubled. The CEO of the company wants to determine whether fixing the bug was a cause to increasing the sales or not to decide if the developer deserve a raise. And if it was a cause, we want to measure and quantify the impact of fixing the bug to the sales. Now there may be other factors (Z) that may have affected the number of orders including the marketing campaign, the number of restaurants, etc. The CEO can see that there is a correlation between the bug fixing and number of orders, but he understands that correlation doesn't imply causation.

So we want to block the impact of other factors Z on the outcome (No. orders) so the only impact on the outcome comes from the treatment (Fixing bug) we are studying. And here's where back-door adjustment comes to place, backdoor adjustment is used to adjust for these confounders (Z) so we can easily measure the impact of the treatment. By adjusting for the confounder variables we mean including them as covariates and conditioning on these variables. See figure 3 which includes all three types of variables in a causal model (treatment, outcome and confounding variables).

However, we can't include all observed variables as covariates, as this will increase the number of variables which may cause over-fitting, also some variables may be hard to measure. Back-door adjustment solves this problem as it helps identifying back-door variables and blocking all back-door paths. A back-door variable is the variable that is correlated

Fig. 3: Example of types of variables in a causal model



with the outcome and the treatment, so it forms a backdoor path of correlation between the treatment and the outcome in which non-causal association flows. That's why we need to block all back-door paths when we only want to measure the causal association between variables. So, we use the back-door adjustment algorithm to identify the variables we need to consider as covariates (condition on) to isolate the impact of the treatment on the outcome. The back-door adjustment is defined as:

Given an ordered pair of variables (X, Y) in a directed acyclic graph G, a set of variables Z satisfies the backdoor criterion relative to (X, Y) if no node in Z is a descendant of X, and Z blocks every path between X and Y that contains an arrow into X. [1]

This simply means we want to block any and all spurious paths while keeping direct paths between treatment and outcome intact. The reason we don't include any descendant of X because we don't want to condition on a collider, A "collider" is a variable that's a descendant to X (treatment) which is correlated with both the exposure and the outcome, and which can therefore produce a spurious relationship between the exposure and the outcome. Conditioning on a collider will unblock a non-causal path between the variables and non-causal association can flow through it, this is what is called collider bias. The intuition behind the previous statement is: by including a variable in the covariates that is caused by both treatment and outcome this may underestimate or overestimate the impact of the treatment itself on the outcome. So, it's crucial not include colliders as covariates when trying to estimate the direct impact between two variables.

Backdoor adjustment algorithm consist of the following steps:

1) Loop over every possible adjustment set (Z)
2) Generate all backdoor paths between T (treatment) and Y (outcome)
3) We then check if all backdoor paths is blocked by conditioning on Z to test the back-door criterion. A path is blocked by two means:
   - Path contains a collider and collider is not in Z.

- Path contains a chain or fork and middle variable is in Z.

4) If the backdoor criterion is satisfied for set Z, then Z is a valid adjustment set.
5) We then use the minimum adjustment set to block all backdoor paths between treatment and the outcome.

The complexity of the previously described algorithm is $O(w!)$ where w is the set of possible adjustment variables. After finding the minimum adjustment set Z that satisfies the backdoor criterion, we can estimate the causal effect of X on Y using the following formula:

$$\mathbb{P}(Y \mid \text{do}(X)) = \sum_z \mathbb{P}(Y \mid X, Z)\mathbb{P}(Z) \tag{1}$$

One of the main limitations the back-door adjustment have is that it assumes all confounding variables are observed, which is not satisfied in some situations. If we have an unobserved (can't be directly measured) confounder, using the back-door adjustment won't be the best solution, it's better to use the front-door adjustment in these cases.

B. Estimation

In causal inference, estimation refers to the process of estimating the causal effect of an exposure or treatment on an outcome based on data from a study or experiment. The goal of causal inference is to determine the extent to which the exposure is responsible for any observed differences in the outcome between the treatment and control groups. There are several different methods for estimating the causal effect of an exposure on an outcome, including:

- Randomization-based methods: These methods rely on the random assignment of individuals to treatment or control groups in order to estimate the causal effect of the exposure on the outcome. Examples include randomized controlled trials (RCTs) and randomized quasi-experiments.
- Observational methods: These methods use data from observational studies, in which individuals are not randomly assigned to treatment or control groups, to estimate the causal effect of the exposure on the outcome. Examples include propensity score matching and instrumental variables analysis.

In this report, we will explore a few observational methods for estimating the causal effect of an exposure on the observed differences in outcome between treatment and control groups. In causal inference and data mining, estimation is the process of assigning a score to the accuracy of the estimand. The conditional average treatment effect (CATE) is a measure of the average effect of an exposure or treatment on an outcome, conditional on the values of one or more covariates. It represents the average treatment effect of the outcome for a specific subgroup of the population, defined by the values of the covariates, and is assumed to be unconfounded and positive. Mathematically, the CATE can be represented as:

$$\tau(x) = \mathbb{E}[Y(1) - Y(0)|X = x] \tag{2}$$

where $\tau(x)$ is the CATE, $Y(1)$ is the outcome for individuals in the treatment group, $Y(0)$ is the outcome for individuals in the control group, and $X = x$ represents a sufficient adjustment set, a subgroup defined by the values of the covariates. The CATE can be estimated using data from an RCT or observational study. In an RCT, the CATE can be estimated by comparing the outcomes of the treatment group and the control group, after adjusting for the effects of the covariates. In an observational study, the CATE can be estimated using methods such as propensity score matching or instrumental variables analysis, which aim to control for the effects of the covariates on the outcome.

The CATE can also be represented as:

$$\begin{aligned}\tau(x) &= \mathbb{E}[Y(1) - Y(0)|X = x] \\ &= \mathbb{E}\, w[\mathbb{E}[Y|T = 1, w] - \mathbb{E}[Y|T = 0, w]] \\ &= \mathbb{E}\, w[\mathbb{E}[Y|T = 1, X = x, w] - \mathbb{E}[Y|T = 0, X = x, w]]\end{aligned}$$

where the last equation holds given that $X \cup W$ is a sufficient adjustment set.

Conditional Outcome Modeling (COM) is a method for estimating the average treatment effect (ATE) of an exposure or treatment on an outcome, using data from a randomized controlled trial (RCT) or observational study. Specifically, COM estimates the ATE by fitting separate regression models for the treatment and control groups and then comparing the predicted outcomes given the values of the confounders. This is expressed mathematically as:

$$\tau = ATE = \mathbb{E}[Y(1)|X] - \mathbb{E}[Y(0)|X] \tag{3}$$

Where ATE is the average treatment effect, $Y(1)$ is the outcome for individuals in the treatment group, $Y(0)$ is the outcome for individuals in the control group, and X represents the confounders.

COM is an observational method that does not rely on random assignment of individuals to treatment or control groups. It is useful for estimating the causal effect of an exposure on an outcome when data from an RCT is not available or is not practical.

It is important to carefully consider the assumptions and limitations of COM when using this method for causal inference. COM relies on the assumption that the confounders have been correctly identified and adequately controlled for in the analysis.

The adjustment formula for COM estimates the ATE by modeling the outcome as a function of the treatment and confounders, and then estimating the difference between the predicted outcomes for the treatment and control groups, given the values of the confounders. This allows COM to adjust for the effects of the confounders on the outcome, and estimate the ATE more accurately. The adjustment formula can be represented as

$$\tau = \mathbb{E}\left[\mathbb{E}[y|T = 1, w] - \mathbb{E}[Y|T = 0, w]\right] \tag{4}$$

In data mining terms, the $\mathbb{E}[Y|a, b]$ is a model of Y conditioned on both a and b. The modeling part can be viewed as a

stratification process in which the data is being stratified and adjusted to ensure that the effect is only a result of the variable being observed (i.e., blocking the path in graphical terms). The expectation over the models represents the matching operation, which shows the effect of exposure on the outcome.

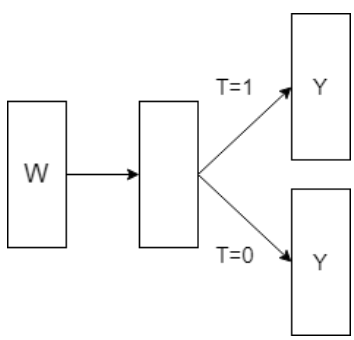
One potential issue with using the COM estimator when the vector of confounders w is high-dimensional is that the model may assign a small weight to the treatment, resulting in a small influence on the outcome Y. A solution to this problem is to use separate models for each treatment value, $T = 0$ and $T = 1$, such that:

$$\tau = \mathbb{E}\left[\mathop{\mathbb{E}}_{T=1}[Y|w] - \mathop{\mathbb{E}}_{T=0}[Y|w]\right] \tag{5}$$

This approach, often referred to as generalized COM (GCOM), involves training one model on data from the treatment group and another model on data from the control group. However, using separate models for each treatment value can also lead to higher variance in the estimator compared to using a single model, as in the traditional COM approach.

To address both issues, the TARNet [2] (Treatment-aware Regression Network) approach involves training a base model on the entire dataset (containing both treatment and control groups) and using separate model estimators to estimate $\tau_{T=1}$ for the treatment group and $\tau_{T=0}$ for the control group. This is achieved by using a neural network with a shared body and two separate heads, one for each treatment value. Figure 4 shows a possible architecture of TARNet.

Fig. 4: Tarnet Architecture



IV. Bounds and Sensitivity Analysis

A. Bounds

In the "Why" library, we have implemented several methods for bound estimation of the average treatment effect (ATE). These bounds provide a range within which the ATE is likely to lie, assuming certain conditions hold. For all of the following bounds, $\pi = P(T = 1)$. Also, to simplify the equations, assume the outcome has two values only a and b.

No assumptions bounds: These bounds do not make any assumptions about the treatment assignment process or the relationships between the variables. They are calculated as:

$$LB = \pi\,\mathbb{E}[Y|T=1] + (1-\pi)b - \pi a - (1-\pi)\,\mathbb{E}[Y|T=0]$$
$$UB = \pi\,\mathbb{E}[Y|T=1] + (1-\pi)a - \pi b - (1-\pi)\,\mathbb{E}[Y|T=0]$$

Monotone treatment response: These bounds assume that the treatment has a monotonically increasing effect on the outcome. Formally, for the nonnegative, we assume for all i, $Y_i(1) >= Y_i(0)$. In that case, we can lower bound the ATE by zero. In the opposite nonpositive case, we can upper bound the ATE by zero.

Monotone treatment selection: These bounds assume that the treatment assignment process is monotone in the sense that the probability of receiving treatment increases with the value of the treatment variable. Formally, we assume $\mathbb{E}[Y(1)|T = 1] >= \mathbb{E}[Y(1)|T = 0]$ and $\mathbb{E}[Y(0)|T = 1] >= \mathbb{E}[Y(0)|T = 0]$. This gives us the following upper bound:

$$UB = \mathbb{E}[Y|T=1] - \mathbb{E}[Y|T=0]$$

Optimal treatment selection: These bounds assume that the treatment assignment process is optimal in the sense that it maximizes the difference between the treated and control groups. Formally, we assume that if $T_i = 1$, then $Y_i(1) >= Y_i(0)$ and if $T_i = 0$ then $Y_i(0) > Y_i(1)$. These assumptions leave to two different upper and lower bounds from which the final bounds are greedily chosen:

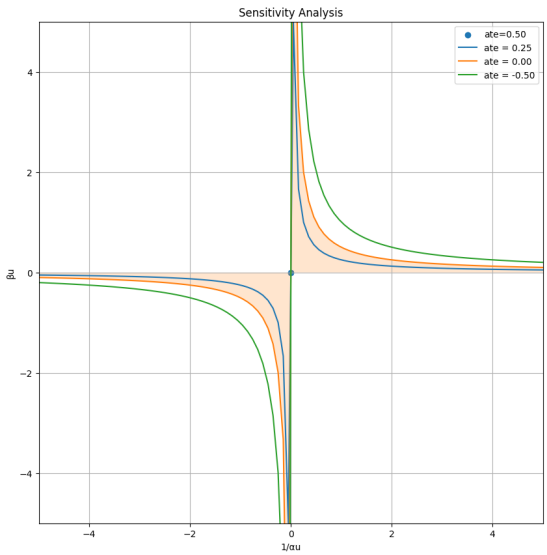
$$LB1 = (1-\pi)a - (1-\pi)\,\mathbb{E}[Y|T=0]$$
$$UB1 = \pi\,\mathbb{E}[Y|T=1] - \pi a$$
$$LB2 = \pi\,\mathbb{E}[Y|T=1] + (1-\pi)a - (1-\pi)\,\mathbb{E}[Y|T=0]$$
$$UB2 = \mathbb{E}[Y|T=1] + \pi a - \mathbb{E}[Y|T=0]\,\mathbb{E}[Y|T=0]$$

B. Sensitivity Analysis

In the "Why" library, we have implemented sensitivity analysis for the presence of an unobserved linear confounder. The purpose of sensitivity analysis is to evaluate the robustness of causal estimates to the presence of unobserved confounding variables. In particular, we consider the case of a single linear unobserved confounder, which is a variable that is correlated with both the treatment and the outcome, and follows a linear relationship with both.

To perform sensitivity analysis, we generate a plot showing the range of possible causal estimates given various assumptions about the strength of the unobserved confounder. In the plot, the solid line represent contour plots where the ATE can be certain values given certain levels of confounding. The point in the center is estimate of the causal effect, which is the estimate obtained without considering the presence of an unobserved confounder. The shaded region represents the range of possible causal estimates given the uncertainty about the strength of the unobserved confounder. If this region is narrow, it indicates that the causal estimate is not robust to the presence of the unobserved confounder. If the region is wide, it indicates that the causal estimate is robust to the unobserved confounder. By examining the sensitivity analysis plot in figure 5, one can assess the robustness of the causal estimate to the presence of an unobserved linear confounder, and make a more informed decision about the validity of the causal inference.

Fig. 5: Sensitivity Analysis Plot

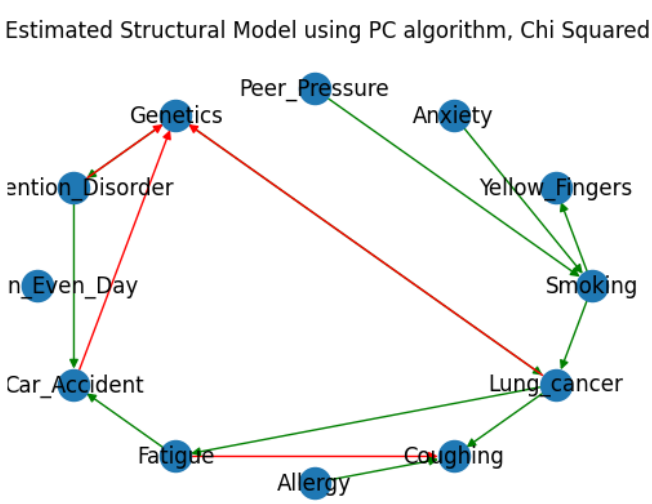


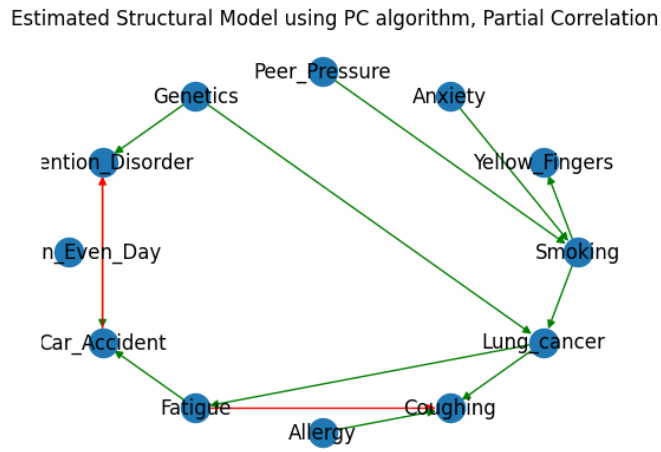
V. Experiments

A. Lucas Dataset

The Lung Cancer Simple (LUCAS) [3] dataset is a dataset used for causal inference research. It consists of data collected from patients with lung cancer, including variables such as age, gender, smoking status, and various clinical measurements. The dataset has been used in research on causal inference methods, particularly in the development and evaluation of methods for estimating causal effects from observational data. It is a relatively small dataset, with only a few hundred observations, but has proven useful for testing and comparing the performance of different causal inference techniques.

1) Causal Graph Construction: We provide the results of the PC Algorithm using the two implemented independence tests: Chi-squared and Partial Correlation tests





where the red arrows represent the missing and or added correlations compared the ground truth. The accuracy of the results is given in the following table:

Method	Accuracy
Partial Correlation	84.6154%
Chi Squared	73.3333%

2) Back-door adjustment: Our back-door adjustment implementation was able to identify all valid adjustment sets, and we used the minimum adjustment set which was an empty set in the case of the lucas dataset.

3) Estimation: The following are the results estimating the average treatment effect (ATE) using COM, GCOM, and TARNet with different models.

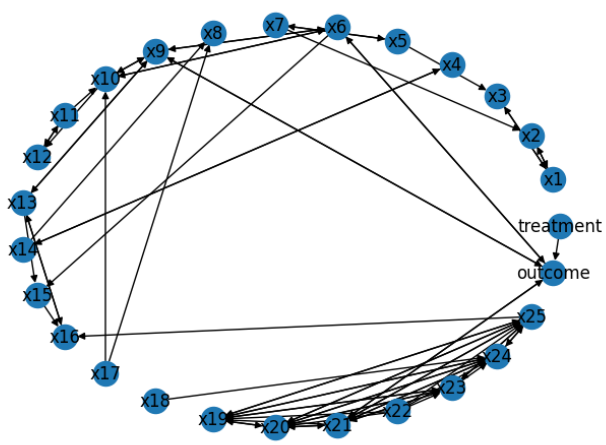
Method	Estimated ATE	Adjustment
COM (Logistic)	0.5041	No
COM (SVM)	0.5064	No
GCOM (Logistic)	0.5105	No
GCOM (SVM)	0.5105	No
GCOM (Logistic)	0.5131	Yes
TarNet	0.5105	No
TARNet	0.5104	Yes

TABLE I: Evaluation of the Causal Estimators

B. IHDP Dataset

The Infant Health and Development Program (IHDP) dataset is a dataset used for causal inference research in many papers such as [4]. It consists of data collected from a randomized controlled trial called the Infant Health and Development Program, which was conducted in the 1980s to study the effects of early intervention on the cognitive and socioemotional development of low birth weight, premature infants. The dataset includes a variety of variables such as measures of the infants' development at various ages, demographic information about the infants and their families, and information about the interventions received. It has been widely used in research on causal inference and has served as a benchmark for evaluating the performance of various causal inference methods.

1) Causal Graph Construction: We provide the results of the PC Algorithm using Partial Correlation:



2) Back-door adjustment: Our back-door adjustment implementation was able to identify all valid adjustment sets, and we used the minimum adjustment set which was an empty set in the case of the ihdp dataset. However we also used other valid adjustment sets to test the rest of the modules. As long as the adjustment set is valid according to the back-door criterion, the measured impact shouldn't change.

3) Estimation: The following are the results estimating the average treatment effect (ATE) using COM, GCOM, and TARNet with different models.

Method	Estimated ATE	Absolute Error
COM Linear (min adj set)	4.0814	0.0306
COM Linear (max adj set)	3.8994	0.1515
GCOM Linear (min adj set)	4.0814	0.0306
GCOM Linear (max adj set)	4.0568	0.0060
GCOM FCN (max adj set)	4.0233	0.0275
TARNet (max adj set)	4.0812	0.0304

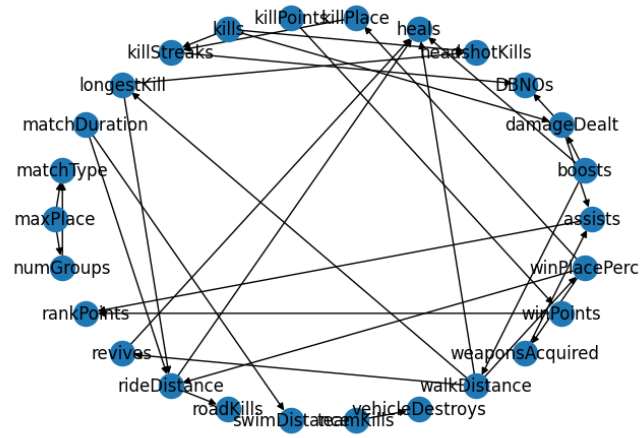
TABLE II: Evaluation of the Causal Estimators

C. PUBG Dataset

The PUBG (PlayerUnknown's Battlegrounds) [5] dataset is a collection of data that is related to the popular battle royale game, PUBG. The dataset might include a variety of information, such as player statistics, game logs, and other data that is collected during gameplay.

In the context of the PUBG dataset, causal inference could be used to understand how different factors within the game might influence player performance or outcomes. For example, the dataset might include information about a player's in-game decisions, their use of weapons or strategies, and their overall success in the game. By analyzing this data, it might be possible to determine which factors are most important in determining player performance and to identify relationships between these factors and player outcomes.

1) Causal Graph Construction: We provide the results of the PC Algorithm using Partial Correlation:



We estimated a few causal effects for fun using a GCOM estimator with a Linear Regression base and obtained the following:

- ATE between kills and winPlacePerc is 0.256
- ATE between damageDealt and assists is 0.227

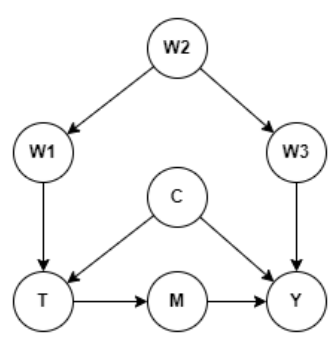
D. Testing the backdoor adjuster

We have tested the implemented backdoor algorithm on different causal models, in the following section we will list some of the test cases and the results we achieved.

See figure 6 for the first test case, this is a simple causal model, in which all variables are observed and measured. The following are some test adjustment sets we asked the back-door adjuster to evaluate whether they were a valid adjustment set or not:

- is the empty set $\{\}$ valid adjustment set. Output: False.
- is $\{C\}$ a valid adjustment set. Output: False.
- is $\{C, W1\}$ a valid adjustment set. Output: True.
- is $\{C, W1, W3\}$ a valid adjustment set. Output: True.
- is $\{W1, W3\}$ a valid adjustment set. Output: False.
- get all valid adjustment sets between T and Y. Output: $\{W2, C\}$, $\{W1, C\}$, $\{C, W3\}$, $\{W2, W1, C\}$, $\{W2, C, W3\}$, $\{W1, C, W3\}$, $\{W2, W1, C, W3\}$.

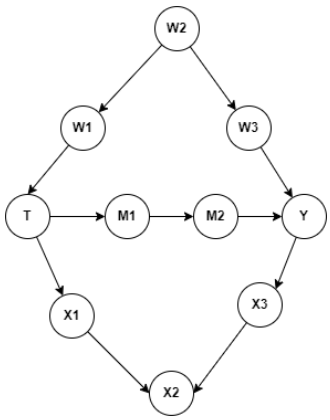
Fig. 6: Back-door adjustment test case 1



See figure 7 for the second test case, this is a more complex causal model than the previous example as the treatment and outcome has a collider, so it's crucial for our implementation to identify the collider (X2) and not output any adjustment sets that include the collider as a covariate . The following are some test adjustment sets we asked the back-door adjuster to evaluate whether they were a valid adjustment set or not:

- is the empty set $\{\}$ valid adjustment set. Output: False.
- is $\{M1\}$ a valid adjustment set. Output: False.
- is $\{M1, W2\}$ a valid adjustment set. Output: True.
- is $\{W1, W3\}$ a valid adjustment set. Output: True.
- is the collider $\{X2\}$ a valid adjustment set. Output: False.
- get all valid adjustment sets. Output: $\{W2\}$, $\{W1\}$, $\{W3\}$, $\{W2, W1\}$, $\{W2, W3\}$, $\{W1, W3\}$, $\{W2, W1, W3\}$.

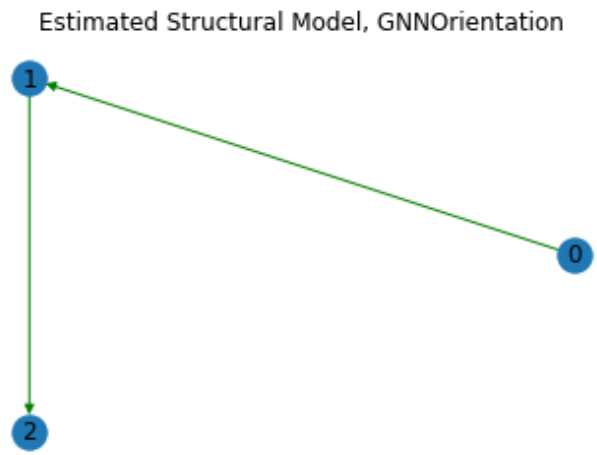
Fig. 7: Back-door adjustment test case 2



E. Testing GNN & CGNN

The GNN and CGNNs were tested in a setup similar to the first experiment in [6]. To evaluate the performance of the CGNNs implemented in the "Why" library, we conducted preliminary experiments using a dataset generated from a linear structural equation model with additive uniform noise. The dataset consisted of 1,500 samples, with the variable X2 being a function of X1, and variable X3 being a function of X2. The CGNNs associated with the relationships $X1 -> X2$ and $X2 -> X1$ were trained using Adam optimization with a learning rate of 0.01, and were evaluated over 500 generated samples, similarly for the X2 and X3 pair. Figure 8 shows the correctly predicted edges by GNN in green. We ran the same experiment using CGNN by starting with a completely incorrect graph where all edges are oriented incorrectly; we achieved the same result as with GNN.

Fig. 8: GNN Orientation Results



VI. Conclusion

In this work, we have presented the "Why" library, a collection of methods for causal inference in observational studies. The library includes methods for causal discovery using the pc algorithm and various independence tests, bound estimation for average treatment effects using different assumptions, sensitivity analysis assuming the presence of an unobserved linear confounder, backdoor adjustment for identifying adjustment sets, and causal effect estimation using COM, GCOM, and TARNet.

In conclusion, the "Why" library provides a comprehensive set of tools for causal inference in observational studies. The library includes a range of methods that can be applied to a variety of settings and data types, and is a valuable resource for researchers and practitioners working in the field of causal inference.

Future work on the "Why" library will focus on expanding the range of methods available and improving the performance and scalability of the existing methods. We also plan to investigate new applications of the "Why" library, such as causal inference in time series data and causal inference with non-linear relationships.

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